



Special issue: Research report

Differences in cytoarchitecture of Broca's region between human, ape and macaque brains

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ARTICLE INFO

Article history:

Received 9 May 2018

Reviewed 12 July 2018

Revised 15 August 2018

Accepted 10 September 2018

Published online 24 September 2018

Keywords:

Broca's region

Cytoarchitecture

Evolution

Neuropil

Primate species

ABSTRACT

Areas 44 and 45 have been identified in non-human primates as homologs of the human Broca region. Distribution of large and smaller pyramids and the ventro-lateral localization in the posterior frontal lobe enable their identification in non-human primates. Since only humans hold the ability of language, it has been hypothesized that differences in microstructure may, together with other anatomical factors, e.g., white matter tract connectivity, volumes of cortical areas and their molecular differentiation, be responsible for the lack (non-human primates) or ability (humans) of language. We sought to identify microstructural differences, by quantitatively studying the cytoarchitecture of areas 44 and 45 using layer-specific grey level indices (volume proportion of neuropil and cell bodies) in serially sectioned and cell body stained human, bonobo, chimpanzee, gorilla, orangutan and *Macaca fascicularis* brains. The main results are the interspecies differences in neuropil volume relative to cell bodies in all layers of both areas which allows a grouping of the different species into three major groups: *Homo sapiens* has the largest, great apes a markedly lower, and macaque the lowest neuropil volume. This indicates considerably more space for local and interregional connectivity in human brains, which matches recent studies of fiber tracts and spacing of cortical minicolumns because increasing connectivity also requires more space for axons and dendrites in the neuropil. The evolutionary enlargement of neuropil is, therefore, a major structural difference between humans and non-human primates which may correspond to the underlying functional differences.

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1. Introduction

A comparison of language properties, e.g., syntactic abilities, recursive structure, between human, apes and macaques

shows differences indicative of a considerable evolution in the human lineage (Fitch, 2011, 2018; Fitch, Huber, & Bugnyar, 2010; Friederici, 2017a, b; Friederici, Bahlmann, Heim, Schubotz, & Anwander, 2006a; Hauser, Chomsky, & Fitch, 2002, 2014). Additionally, most available findings on

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<https://doi.org/10.1016/j.cortex.2018.09.008>

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language functions in apes are based on extensively trained paradigms which are far from situations comparable to human language (Rumbaugh, 1977; Savage-Rumbaugh & Rumbaugh, 1979). The FOXP2 transcription factor in apes and monkeys are currently central to genetic analysis of language function (Konopka et al., 2012; Staes et al., 2017), since the amino acid coding and pattern of nucleotide polymorphism in the human FOXP2 gene compared to that of chimpanzee suggest an important role in human language evolution (Enard et al., 2002). However, detailed gene-phenotype mapping to normal functions of both the human and the chimpanzee FOXP2 proteins is required for a better understanding of a gene-driven evolution of language.

Various studies were published on the architectonic properties of cortical areas in the inferior frontal region of non-human primate brains, which are thought to be the homologs of the Broca region in the human brain. Comparisons between the areas of the human Broca region (Broca, 1861) with a similarly located region in non-human primates were performed based on its leftward asymmetry in human, chimpanzee and gorilla brains (Balzeau et al., 2012; Cantalupo & Hopkins, 2001; Keller, Crow, Foundas, Amunts, & Roberts, 2009; Keller, Roberts, Hopkins, 2009), and on macroscopical location (Mingazzini, 1928), cyto- and myeloarchitecture (Brodmann, 1909; Kreht, 1936; Mauss, 1908, 1911; Morecraft, Stilwell-Morecraft, Ge, Cipolloni, & Pandya, 2015; Petrides, Tomaiuolo, Yeterian, & Pandya, 2012; Petrides & Pandya, 1999, 2002; Preuss & Goldman-Rakic, 1991; Rilling, 2014; Schenker, 2007; Schenker et al., 2010, 2008; Semendeferi, 2004; Sherwood et al., 2004, 2003; Sherwood & Hof, 2007; Strassburger 1937; Vogt & Vogt, 1919), as well as connectivity (Frey, Campbell, Pike, & Petrides, 2008; Friederici, 2017a, b; Friederici et al., 2006a; Jbabdi, Lehman, Haber, & Behrens, 2013; Petrides, 2013; Petrides et al., 2012; Petrides & Pandya, 2009; Rilling, 2014; Rilling, Glasser, Jbabdi, Andersson, & Preuss, 2012; Watanabe-Sawaguchi, Kubota, & Arikuni, 1991). A resting state study using PET (Rilling et al., 2007) suggests that chimpanzee brains differ from human brains by a lower activity in left-sided areas homolog to those which are involved in language processing in humans. Other studies identified activation of Broca's region of chimpanzees during the production of attention-getting sounds (Taglialetela, Russell, Schaeffer, & Hopkins, 2011), and homologs of human Wernicke's and Broca's areas associated with perceiving species-specific vocalizations in rhesus monkeys (Gil-da-Costa et al., 2006; Wilson et al., 2015). Thus, at least the Broca homolog regions in chimpanzees are putatively involved in vocal signaling in the common ancestor of both humans and chimpanzees.

The cytoarchitectonic areas 44 and 45 of Broca's region are found in the caudal part of the human inferior frontal gyrus (Brodmann, 1909). This part is subdivided by the highly variable ascending, diagonal, triangular and horizontal sulci and partly bordered by the inferior frontal and precentral sulci (Amunts et al., 1999; Amunts & Zilles, 2012; Petrides, 2013; Sherwood, Broadfield, Holloway, Gannon, & Hof, 2003; Zilles & Amunts, 2018). The latter two sulci are also found in the bonobo, common chimpanzee, gorilla and orangutan brains, but homologs of both sulci are not identifiable in baboon or macaque brains (Eberstaller, 1890; Mingazzini, 1928). These species have instead an arcuate and a principal sulcus which

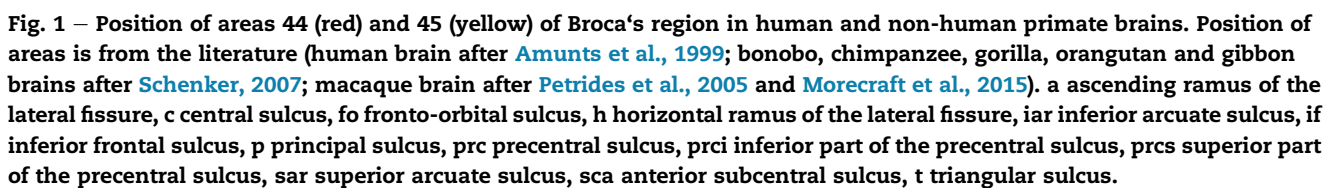
are not present in the human and great ape brains. Since a frontoorbital sulcus is visible in all great ape, baboon and macaque brains with the notable exception of the human brain, this sulcus represents an important difference between non-human primate and human brains. Facing the variability in sulcal pattern of the human brain and the difficulty in identifying homolog sulci in non-human primates, sulci do not predict in each specimen the localization and extent of the cytoarchitectonically defined areas 44 and 45 in human (Amunts et al., 1999) and ape brains (Schenker, 2007; Schenker et al., 2010, 2008; Semendeferi, 2004; Sherwood et al., 2003).

The Broca region of humans and its homologs of non-human primates, particularly great apes, share similar features and maps (Brodmann, 1909; Campbell, 1905; Mauss, 1908, 1911; Petrides et al., 2012, 2005; Petrides & Pandya, 1999; Schenker, 2007; Schenker et al., 2010, 2008; Semendeferi, 2004; Sherwood & Hof, 2007; Strassburger, 1938, 1937). Pyramidal cells in deeper layer III which are larger than those in layer V, a reduced inner granular layer (layer IV) in area 44 and a well recognizable layer IV in area 45. Both areas were found in human, ape and macaque brains in front of the premotor cortex on the inferior part of the frontal lobe (Fig. 1). These criteria emphasize similarities, the language properties, however, are different between humans and non-human primates. Therefore, in the present study we focus on the dissimilarities. Specifically, we identify dissimilarities in cytoarchitecture of Broca's region between human, bonobo, chimpanzee, gorilla, orangutan, and macaque brains.

To define the degree of dissimilarity of the cytoarchitecture of the human Broca region with homolog regions in apes and macaques we quantitatively analyzed the laminar cell body distribution in areas 44 and 45, and measured the volume proportion of the space between cell bodies, since here the dendrites, axons and synapses (structures crucial for information processing) are found.

2. Material and methods

We analyzed four human brains from donors without a history of psychiatric or neurological diseases, obtained from the donor program of the University of Düsseldorf in accordance with legal requirements. Brains were fixed in 4% buffered formalin or Bodian's fixative for at least 6 months. The great ape specimens with YN IDs were donated by the Yerkes National Primate Research Center to Katarina Semendeferi, University of California in San Diego following the natural deaths of the animals. Macaque monkey (*Macaca fascicularis*) brains were obtained from Covance Company, Münster. The macaques were housed in this company and were control animals for pharmaceutical studies performed in compliance with legal requirements. They were sacrificed with thiopentobarbital. Except for human ($n = 4$) and macaque ($n = 3$) brains, each species is represented by a single brain (Table 1) due to the limited availability of serially sectioned ape brains and the necessity to analyze methodically comparable (same staining method) preparations of equal staining quality. Five to 15 sections per brain were examined, depending on the species, and measurements were always carried out in the left hemisphere.



were digitized using a light microscope (Axioplan 2 imaging, ZEISS, Germany) equipped with a motor-operated stage controlled by the KS400® (Zeiss, Germany) image analyzing system (version 3.0) and Axiovision (version 4.6) and with a CCD-camera (Axiocam MRm, ZEISS, Germany). This step provided a digitized data set of the histological sections with an in-plane resolution of 1 μm . For details of the histological processing see (Palomero-Gallagher & Zilles, 2018; Zilles, Schleicher, Palomero-Gallagher, & Amunts, 2002).

Table 1 – Data of specimens for quantitative cytoarchitectonic analysis.

Species	ID	Sex	Age [years]	Brain weight [g]
<i>Homo sapiens</i>	B01	female	79	1350
<i>Homo sapiens</i>	B06	male	54	1622
<i>Homo sapiens</i>	B07	male	37	1437
<i>Homo sapiens</i>	B10	female	85	1046
<i>Pan paniscus</i> (Bonobo)	1/97	female	11	324
<i>Pan troglodytes</i> (Chimpanzee)	YN 89-278	male	22	420
<i>Gorilla gorilla</i> (Gorilla)	YN 82-140	female	20	376
<i>Pongo pygmaeus</i> (Orangutan)	YN 85-38	male	16.5	369
<i>Hylobates lar</i> (Gibbon)	YN 81-146	female	adult	92
<i>Macaca fascicularis</i>	ID11530	male	6	72
(Macaque)	ID11539	male	8	69

The digitized sections were then converted into gray level index (GLI) images (Schleicher et al., 2000). The GLI is a measure of the volume fraction of cell bodies per tissue volume (Schleicher et al., 2005; Schleicher & Zilles, 1990; Wree, Schleicher, & Zilles, 1982). The outer and inner contours of the cortical ribbon were interactively traced under microscopical control. Based on the Laplace equation, traverses running perpendicular to the cortical layers from the outer to the inner traced contours were calculated. Along these traverses, profiles describing the layer-specific changes in the volume fraction of cell bodies (GLI) were extracted throughout the cortical ribbon from the pial surface to the cortex/white matter border. The width of a single traverse was 3 μm , the length depended on the cortical thickness in each area, and was normalized to 100%. Blocks of GLI profiles (each consisting of 11 single profiles) were calculated in each specimen and cortical area. The surface defined beneath the GLI profile, or beneath the discrete sectors defined by the position of borders between layers, was computed to yield the mean GLI for the entire cortical depth or for each layer separately.

Multivariate analyses of the layer-specific GLIs were conducted to visualize putative clusters of species and areas according to the degree of (dis)similarity of their GLI values using the Matlab Statistics Toolbox (MatLab R2009a; Mathworks Inc., Natick, MA, USA) and Systat (Systat 13; Systat Software Inc., Chicago, IL, USA). In these analyses, layer-specific GLI values were treated as feature vectors describing the relation between the volume proportion of neuropil and cell bodies in each layer, area and species. Brain size largely differs between human, ape and macaque brains (Fig. 1). Hierarchical cluster analyses of the GLI values were performed as previously described (Palomero-Gallagher et al., 2009) using the Euclidean distance as a measure of (dis)similarity and the Ward linkage algorithm as the linkage method. The number of clusters in the dendrograms was defined using the k-means clustering and the elbow method. The separation quotients, which constitute the basis for performing the elbow assessment of the number of clusters, were calculated using the Matlab Statistics Toolbox (MatLab R2009a; Mathworks Inc., Natick, MA, USA). Additionally, principal component analyses were carried out.

Finally, the GLI in areas 44 and 45, as well as in the primary motor cortex (area 4), prefrontal area 10, orbitofrontal area 13, and visual areas V1 and V2 were correlated with brain weight to test the possible dependence of GLI on brain weight. Data for areas 4, 10, 13, and V1 and V2 were obtained from the literature (de Sousa et al., 2010; Semendeferi, Armstrong, Schleicher, Zilles, & Van Hoesen, 2001, 1998; Sherwood et al., 2004). Since de Sousa et al. (2010) originally reported grey values obtained from 8-bit images of areas V1 and V2, and not GLI values, we corrected their data to GLI values in the present study to ensure comparability with the present measurements and the other data from the literature. This only resulted in absolute changes in scaling of the y axis, but retained the interspecies differences reported by the authors for these two areas. Furthermore, we measured the GLIs of primary motor cortex (area 4) in bonobo and gibbon brains since these data were not reported by Sherwood et al. (2004).

3. Results

The cytoarchitecture of human areas 44 and 45 is characterized by some very large pyramidal cells or clusters of those in layer IIIc, and large pyramidal cells in layer Va (Figs. 2 and 3). The size of cells in layer III increases from its border with layer II and to the layer III/IV border. The most important difference between areas 44 and 45 is found in layer IV, which is dysgranular and invaded by pyramidal cells of layers III and V in area 44. In contrast, layer IV of area 45 is much wider and only seldom invaded by layer III or layer V pyramids. Areas 44 and 45 could be identified by the same criteria across all ape brains (Figs. 4–8) and the macaque (Fig. 9).

The cytoarchitecture of both areas was quantitatively analyzed in each species using the method of GLI measurement (see Material and Methods) vertical to the cortical surface across all layers. The resulting single GLI profiles are shown in Figs. 2–9. In these figures, the vertical line indicates the precise course from which the GLI profile was extracted. The width of a single profile is 3 μm .

To obtain GLI profiles covering a higher portion of the cortical ribbon than the single profiles, we averaged 11 immediately adjoining profiles covering a total width of 33 μm in both areas of all species. Even after averaging over 33 μm , the GLI courses show considerable fluctuations within and between layers of both areas (Figs. 10 and 11). Further averaging to reduce the variability of the GLI courses in apes and macaques was not performed, since it would introduce an artificial blurring of the laminar pattern due to the varying thickness of the entire width of the cortex and individual layers depending on the actual position along a folded cortical surface (Bok, 1929). These effects of cortical folding on the shape of GLI courses do not occur in a tightly confined measuring field. In all brains of the different species, the lowest values in areas 44 and 45 are found in layers I and VIb. Local maxima are variably localized in the different layers of areas 44 (Figs. 10) and 45 (Fig. 11).

The highest mean (averaged over all cortical layers) GLI is observed in human areas 44 and 45. Only in some cases the chimpanzee reaches comparable values (Fig. 10A). In these areas, the mean GLIs of the other non-human primates are

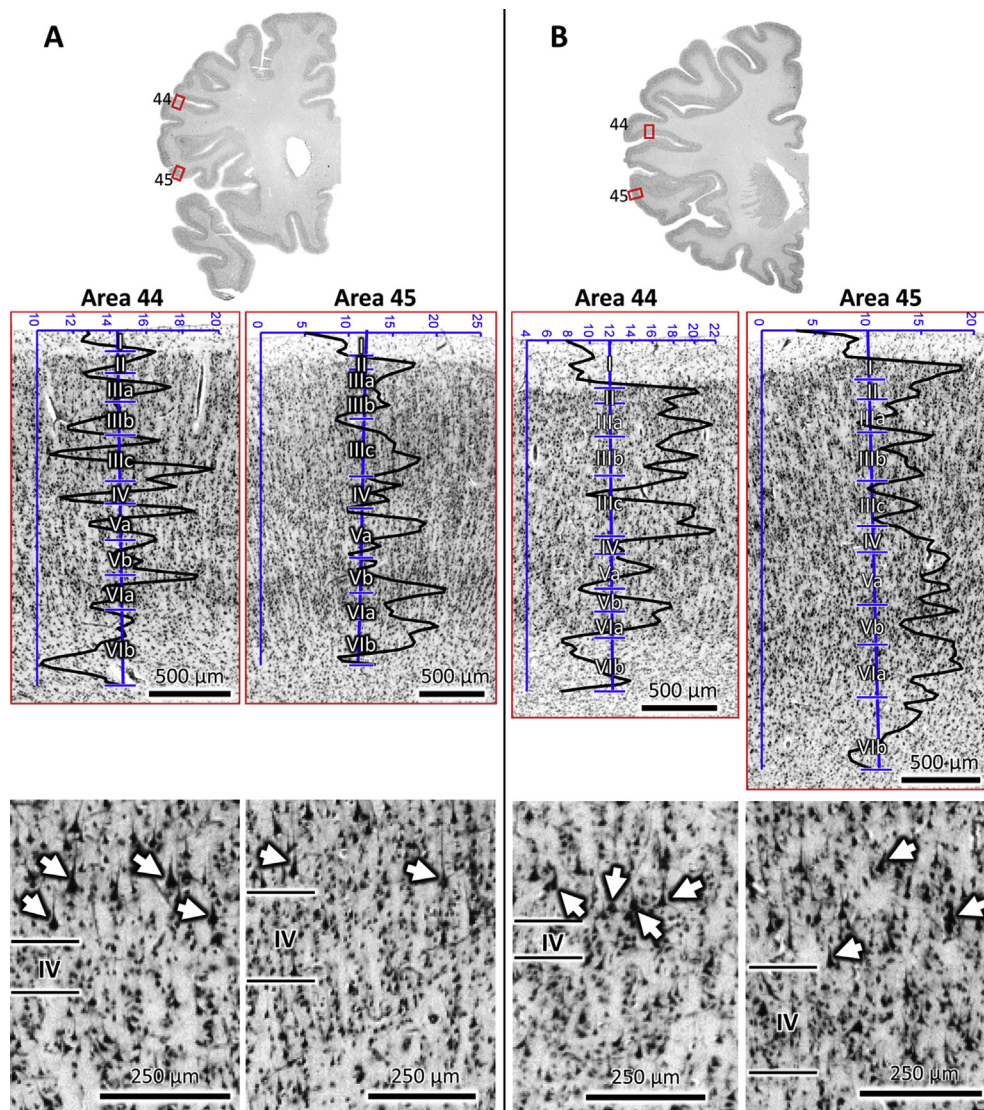


Fig. 2 – Cytoarchitecture of areas 44 and 45 of the Broca region in the human brains B01 (A; section 5296) and B06 (B; section 5176). Top row, coronal sections. Middle row, grey level index profiles (covering a 3 μm wide cortical column) through areas 44 or 45 displayed on magnifications from A or B (red rectangles show the respective positions in the overviews). Bottom row, detailed view of the larger pyramidal cells in layer IIIc (arrowheads) compared to layer V (“externopyramidization”; Goulas, Zilles, & Hilgetag, 2018) of areas 44 and 45. Roman numerals identify cortical layers.

lower than the human GLIs of both areas (Fig. 12). It is important to emphasize that GLI values for the different ape species were obtained from single specimens. Thus, in contrast to the human and macaque data, information concerning inter-individual variability of the cytoarchitecture in apes is not provided.

If the single layers are grouped into three strata with layers II–IIIc as supragranular, layer IV as granular and layers Va–VIb as infragranular strata, the human brain shows the highest GLIs in all three strata in both area 44 and area 45 (Fig. 12B,C). In area 44, the chimpanzee reaches the human values in the supra- and infragranular layers, whereas the bonobo shows a lower GLI in the supragranular, but higher GLIs in the granular and infragranular strata than gorilla, orangutan and macaque (Fig. 12B). Thus, in area 44 the chimpanzee reaches a similar volume proportion of cell bodies compared with the human

brain in the supra- and infragranular strata and nearly the same value in the granular stratum. In area 45, the bonobo shows the lowest GLI in the supragranular stratum, whereas the chimpanzee, gorilla, orangutan and macaque display intermediate values. In the granular and infragranular strata of area 45, bonobo and chimpanzee take an intermediate position between the high GLI of the human and the low GLIs of the gorilla, orangutan and macaque brains (Fig. 12C). Thus, a similar situation as in area 44 is also found in area 45 when comparing the GLIs of the three strata across species.

The very large human brains have the largest mean GLI, and the large ape brains have larger mean GLIs than the small macaque brains. The volume of the white matter also increases with increasing brain/teiencephalic and cortical volume, and the volume of the grey and white matter is correlated in these differentially sized brains. Consequently, it

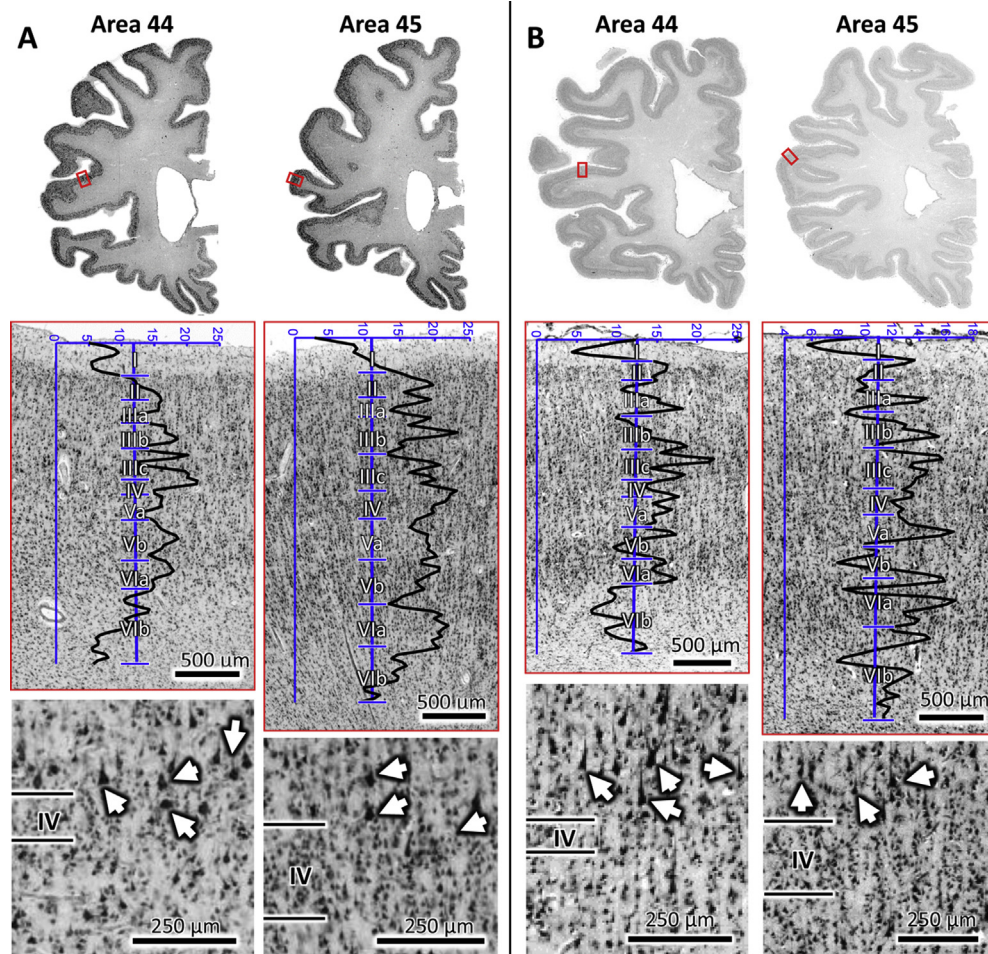


Fig. 3 – Cytoarchitecture of areas 44 and 45 of the Broca region in the human brains B07 [A; sections 5161 (area 44) and 5341 (area 45)] and B10 [B; sections 5311 (area 44) and 5656 (area 45)]. Top row, coronal sections. Middle row, grey level index profiles (covering a 3 μm wide cortical column) through areas 44 or 45 displayed on magnifications from A or B (red rectangles show the respective positions in the overviews). Bottom row, detailed view of the larger pyramidal cells in layer IIIc (arrowheads) compared to layer V (“externopyramidization”; Goulas et al., 2018) of areas 44 and 45. Roman numerals identify cortical layers.

can be assumed that more input and output axons/fibers may be present in the grey matter of brains with larger white matter volumes. This could lead to a lower GLI value in large brains. Fig. 13 shows, however, that these hypotheses are not supported by the data from different brain regions. If we compare the correlations between brain weight and GLI of areas 4, 10, 13, V1 and V2 (data from Semendeferi, Armstrong, Schleicher, Zilles, & Van Hoesen, 1998, 2001; de Sousa et al., 2010; Sherwood et al., 2004; for details see Material and Methods) with our measurements in areas 44 and 45, we can observe opposing significant correlations or non-significant trends. The GLI averaged over all cortical layers, supra-granular, granular and infragranular layers increases with increasing brain weight in areas 44 and 45, whereas a negative correlation is found for areas 4, 10, 13, V1 and V2. Thus, the neuropil proportion increases in the latter areas with increasing size, but decreases in areas 44 and 45.

To answer the initial question regarding quantitative (dis) similarities of the cytoarchitecture between Broca's region in the human brain and its homologs in ape and macaque brains,

hierarchical cluster and principal component analyses of the GLIs normalized by brain size in the supragranular, granular and infragranular layers of areas 44 and 45 were performed over all brains studied here (Fig. 14). In both areas, the cytoarchitecture in the human brain is different from that of all non-human primates. The largest difference in cytoarchitecture of areas 44 and 45 is seen between the human brain on one side, and the gibbon and macaque brains on the other side. Also the bonobo brain is located in an own cluster, and is separated from the remaining great apes in both areas. The gorilla, orangutan and chimpanzee brains are found in different clusters in areas 44 and 45, but the differences in cytoarchitecture are extremely small (Euclidean distances between .005 and .007). These results are further supported by the principal component analyses (Fig. 14B,D).

If we extend the comparisons to areas 4, 10, 13, V1 and V2, we find similar results of the hierarchical cluster and the principal component analyses of the brain-size weighted GLI values (Fig. 15 and Supplementary Fig. 1). Three clusters are found in areas 10 and V1, with the human and all great ape

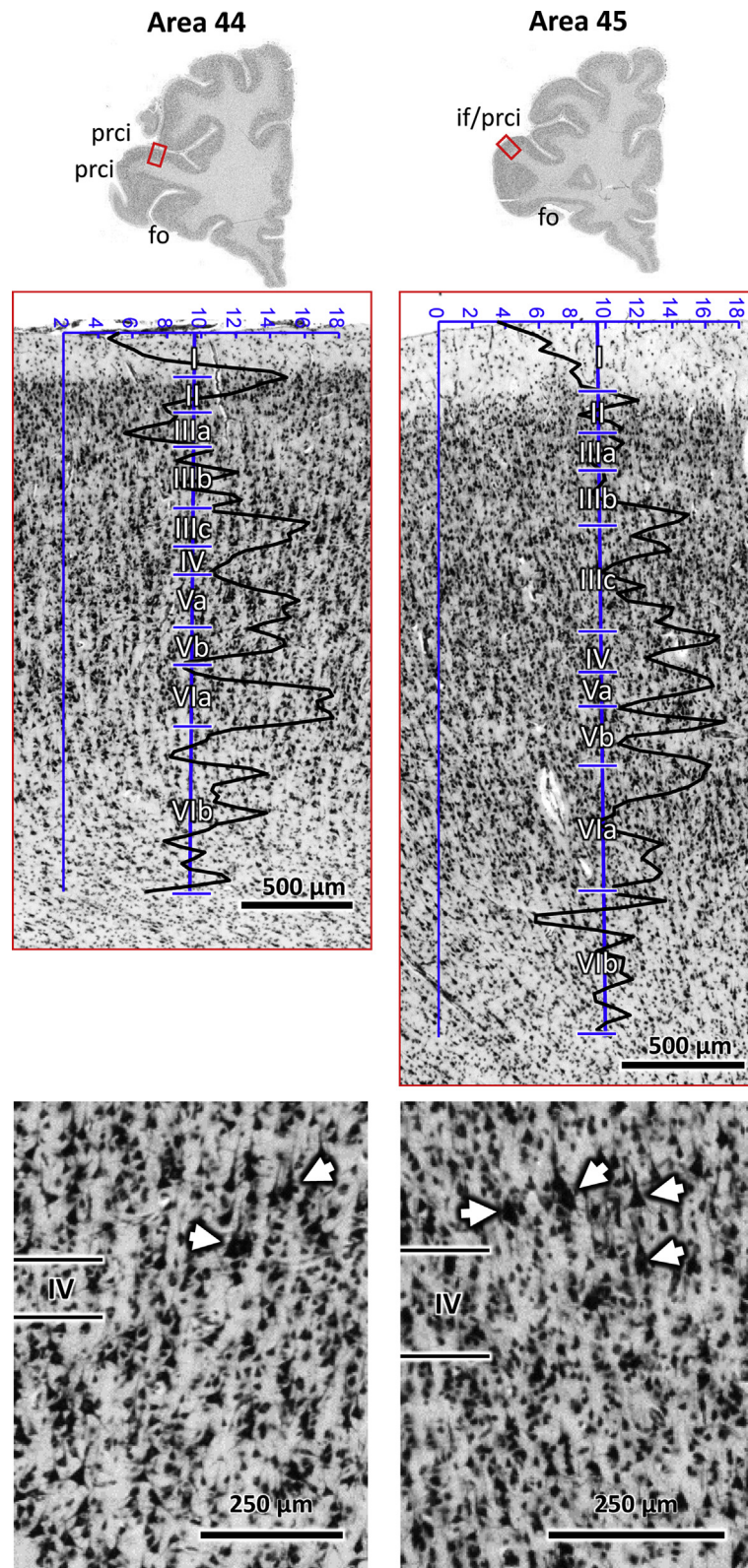


Fig. 4 – Cytoarchitecture of areas 44 and 45 of the Broca region of *Pan paniscus* [1/97; sections 3561 (area 44) and 3711 (area 45)]. Top row, coronal sections. Middle row, grey level index profiles (covering a 3 µm wide cortical column) through areas 44 or 45 displayed on magnifications from A or B (red rectangles show the respective positions in the overviews). Bottom row, detailed view of the larger pyramidal cells in layer IIIc (arrowheads) compared to layer V (“externopyramidization”; Goulas et al., 2018) of areas 44 and 45. fo fronto-orbital sulcus, if/prci junction of the inferior frontal sulcus and the inferior part of the precentral sulcus, prci inferior part of the precentral sulcus. Roman numerals identify cortical layers.

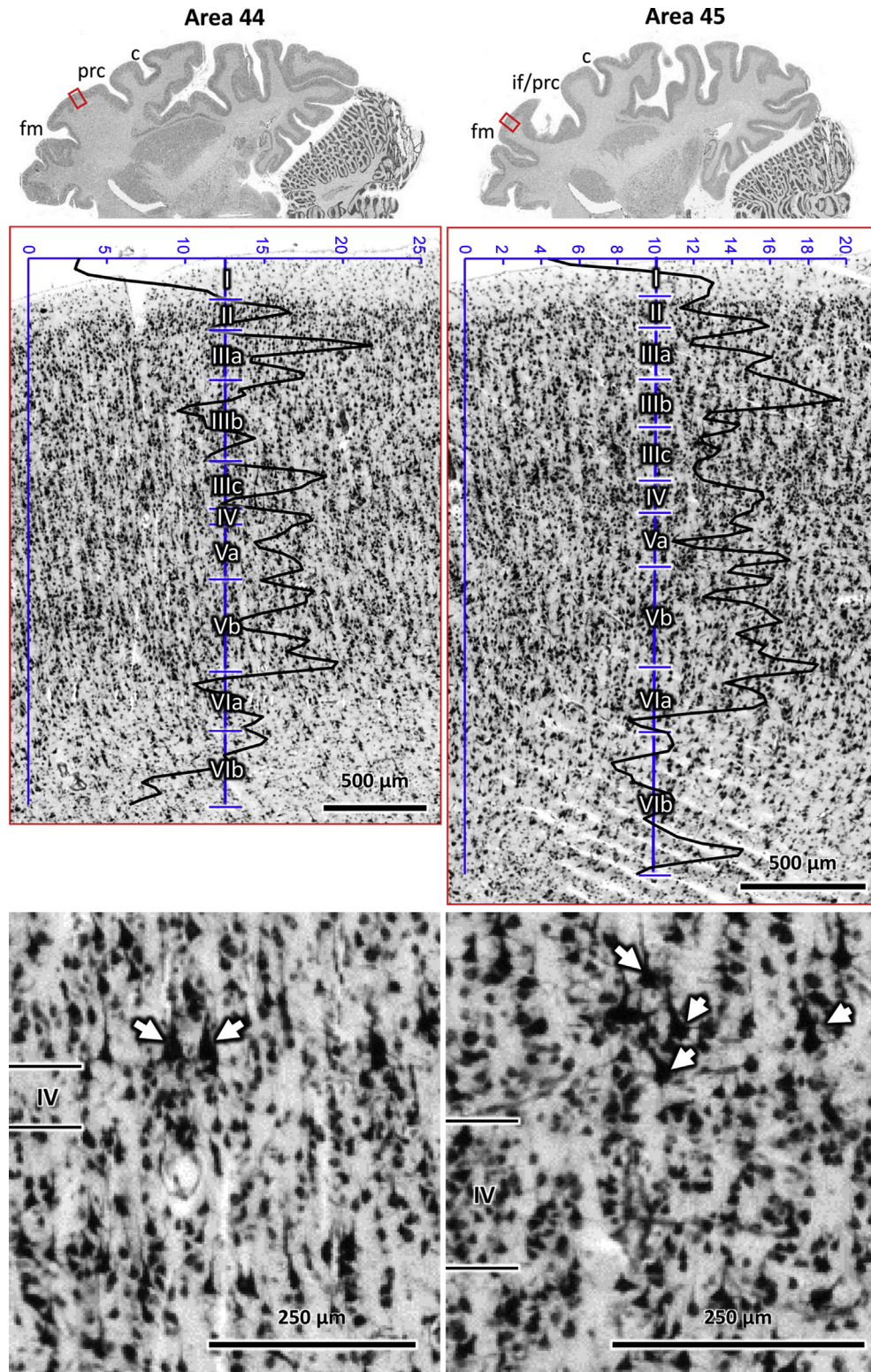


Fig. 5 – Cytoarchitecture of areas 44 and 45 of the Broca region of *Pan troglodytes* [YN 89–278; sections 1381 (area 44) and 1681 (area 45)]. Top row, horizontal sections. Middle row, grey level index profiles (covering a 3 µm wide cortical column) through areas 44 or 45 displayed on magnifications from A or B (red rectangles show the respective positions in the overviews). Bottom row, detailed view of the larger pyramidal cells in layer IIIc (arrowheads) compared to layer V (“externopyramidization”; Goulas et al., 2018) of areas 44 and 45. c central sulcus, fm frontomarginal sulcus, if inferior frontal sulcus, prc precentral sulcus. Roman numerals identify cortical layers.

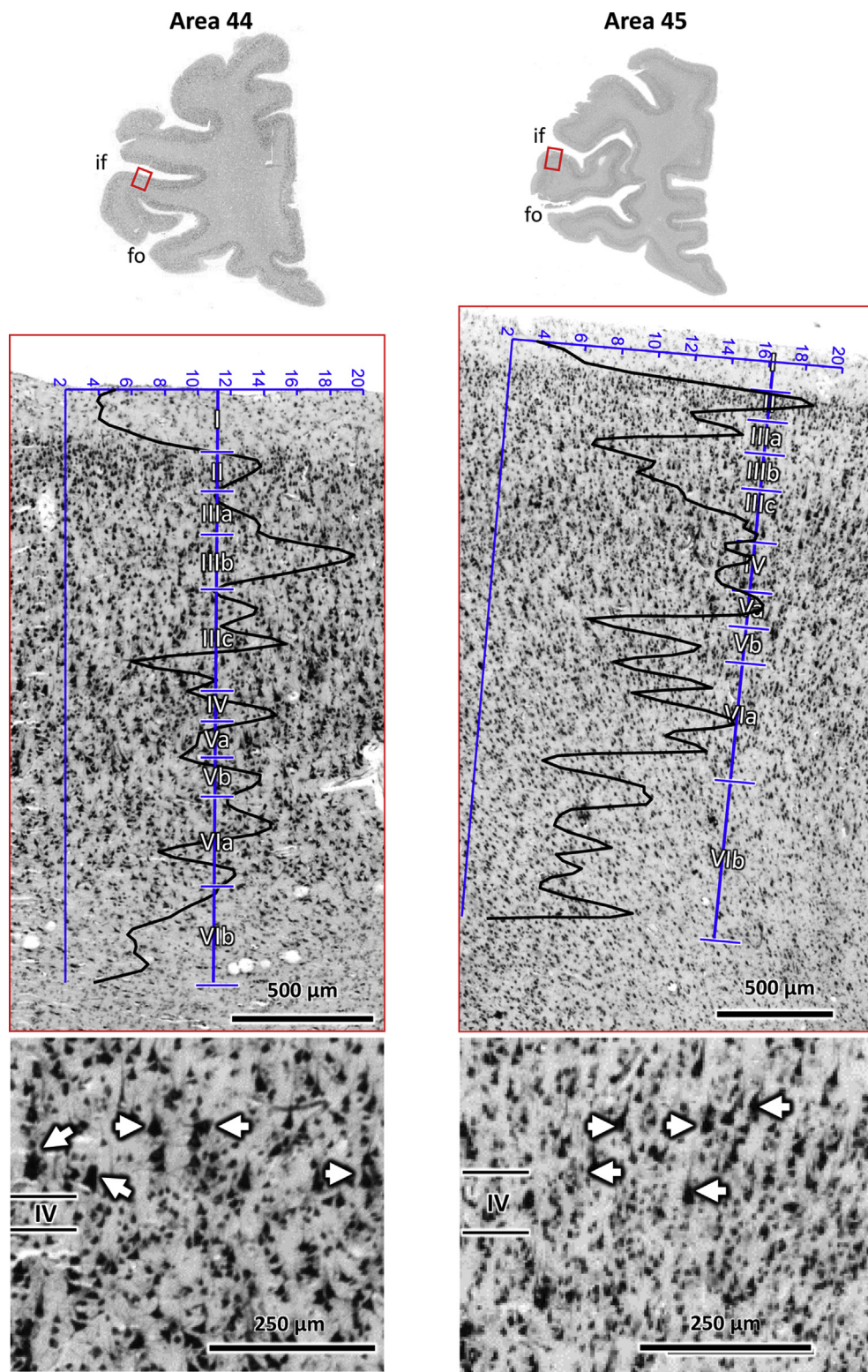


Fig. 6 – Cytoarchitecture of areas 44 and 45 of the Broca region of *Gorilla gorilla* [YN 82–140; sections 3840 (area 44) and 4256 (area 45)]. Top row, coronal sections. Middle row, grey level index profiles (covering a 3 μm wide cortical column) through areas 44 or 45 displayed on magnifications from A or B (red rectangles show the respective positions in the overviews). Bottom row, detailed view of the larger pyramidal cells in layer IIIc (arrowheads) compared to layer V (“externopyramidization”; Goulas et al., 2018) of areas 44 and 45. fo fronto-orbital sulcus, if inferior frontal sulcus. Roman numerals identify cortical layers.

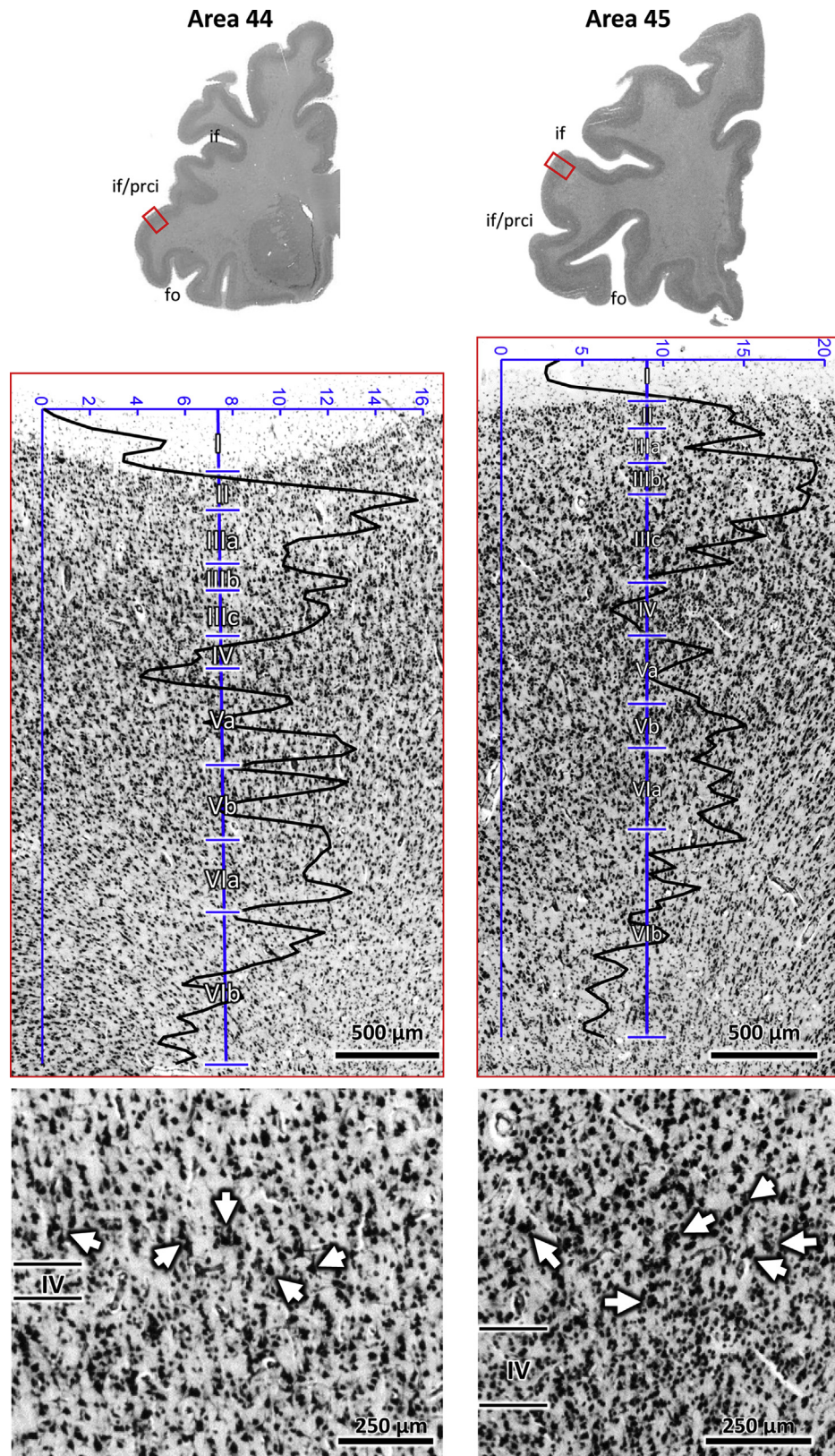


Fig. 7 – Cytoarchitecture of areas 44 and 45 of the Broca region of *Pongo pygmaeus* [YN 85-38; section 3361 (area 44) and section 3541 (area 45)]. Top row, coronal sections. Middle row, grey level index profiles (covering a 3 μm wide cortical column) through areas 44 or 45 displayed on magnifications from A or B (red rectangles show the respective positions in the overviews). Bottom row, detailed view of the larger pyramidal cells in layer IIIc (arrowheads) compared to layer V (“externopyramidization”; Goulas et al., 2018) of areas 44 and 45. fo fronto-orbital sulcus, if inferior frontal sulcus, if/prci junction of the inferior frontal sulcus and the inferior part of the precentral sulcus. Roman numerals identify cortical layers.

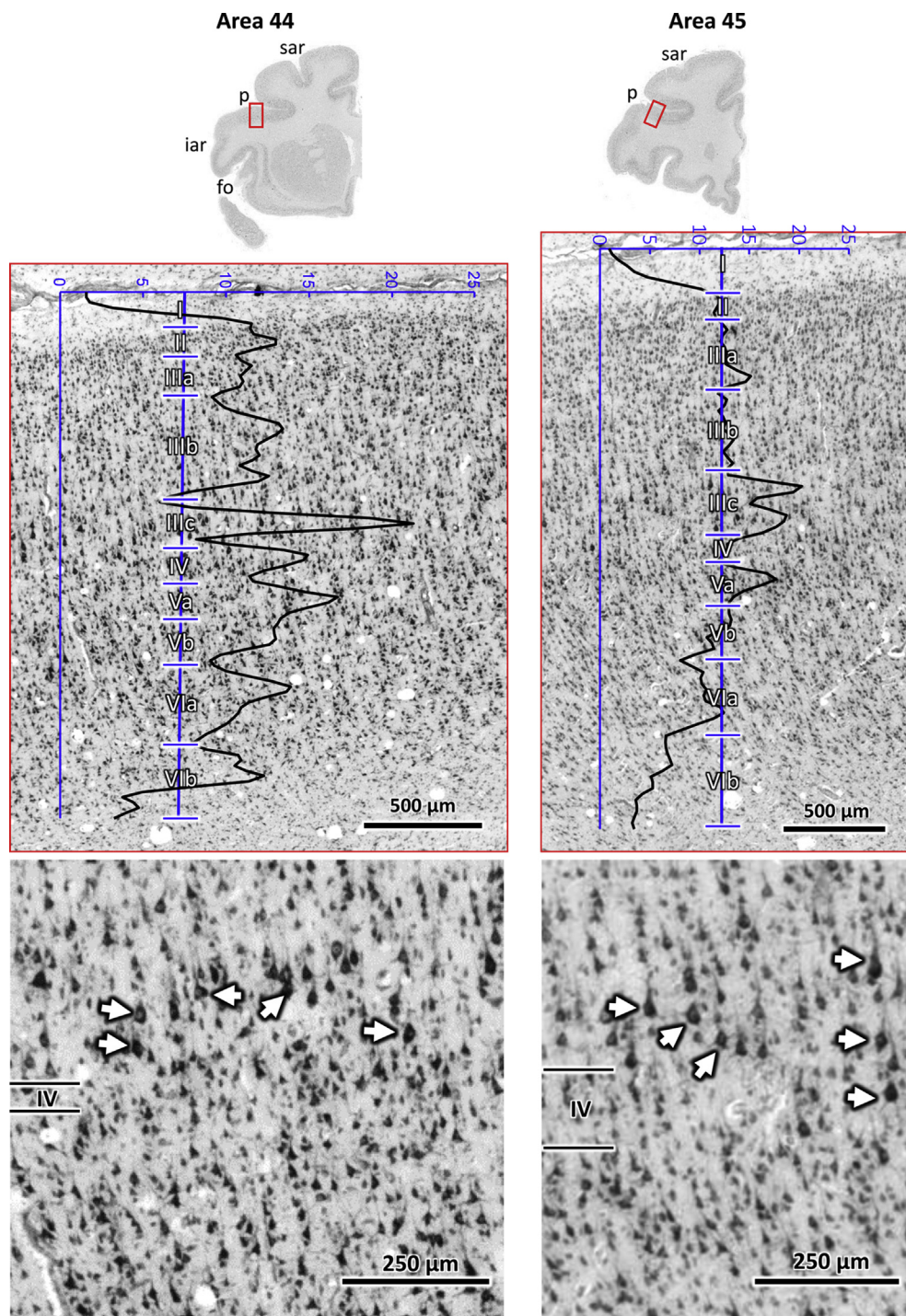


Fig. 8 – Cytoarchitecture of areas 44 and 45 of the Broca region of *Hylobates lar* [YN 81–146; sections 1971 (area 44) and 2161 (area 45)]. Top row, coronal sections. Middle row, grey level index profiles (covering a 3 µm wide cortical column) through areas 44 or 45 displayed on magnifications from A or B (red rectangles show the respective positions in the overviews). Bottom row, detailed view of the larger pyramidal cells in layer IIIc (arrowheads) compared to layer V (“externopyramidization”; Goulas et al., 2018) of areas 44 and 45. iar inferior arcuate sulcus, fo fronto-orbital sulcus, p principal sulcus, sar superior arcuate sulcus. Roman numerals identify cortical layers.

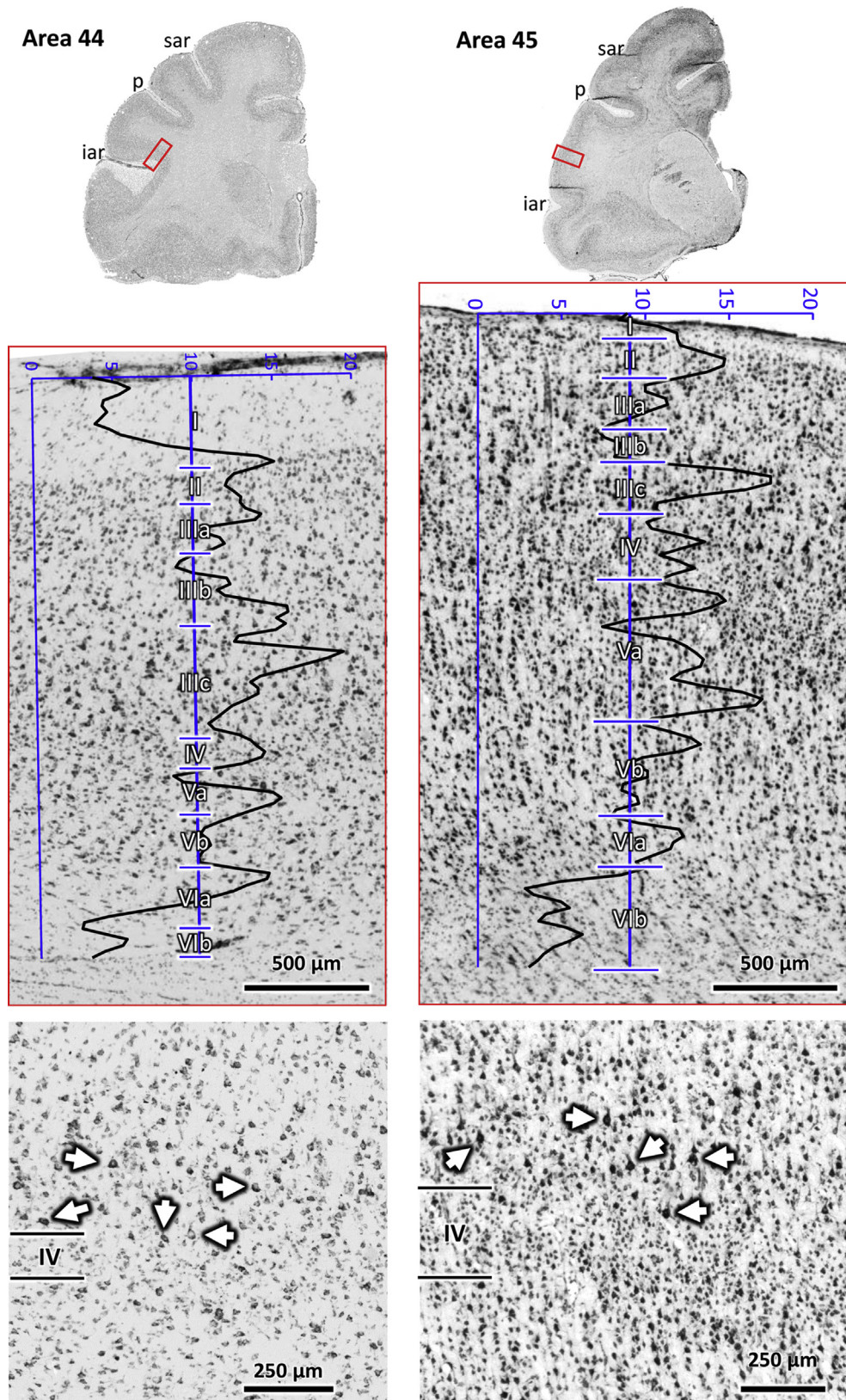


Fig. 9 – Cytoarchitecture of areas 44 and 45 of the Broca region of *Macaca fascicularis* [ID11530; section 53 (area 44) and section 55 (area 45)]. Top row, coronal sections. Middle row, grey level index profiles (covering a 3 μm wide cortical column) through areas 44 or 45 displayed on magnifications from A or B (red rectangles show the respective positions in the overviews). Bottom row, detailed view of the larger pyramidal cells in layer IIIc (arrowheads) compared to layer V (“externopyramidization”; Goulas et al., 2018) of areas 44 and 45. iar inferior arcuate sulcus, ipd inferior principal dimple, p principal sulcus, sar superior arcuate sulcus. Roman numerals identify cortical layers.

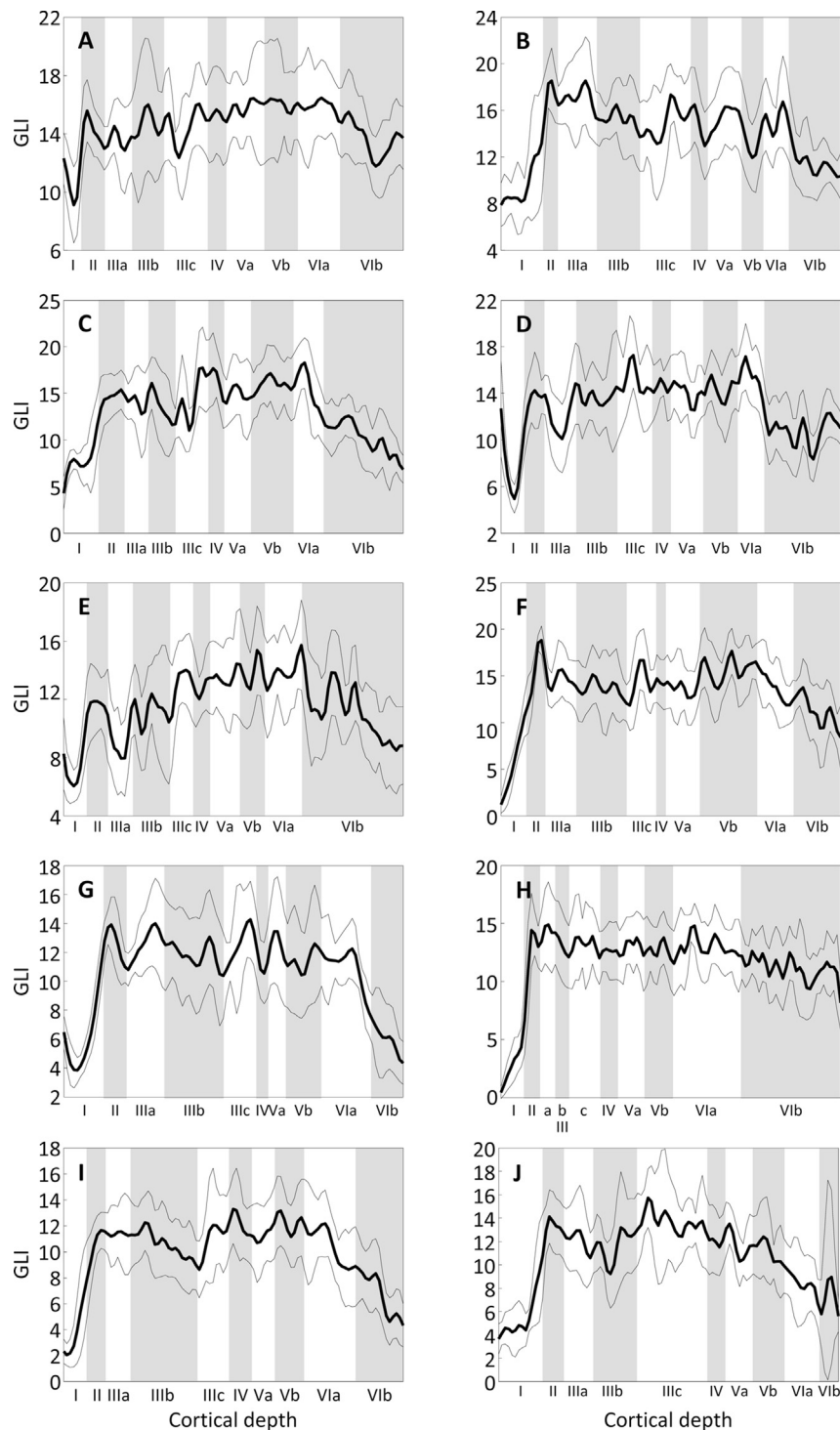


Fig. 10 – Mean GLI profiles of area 44. Eleven adjacent single profiles (each covering a 3 μm wide cortical column; total width 33 μm) were extracted and averaged to obtain the mean GLI profile. A human (B01), B human (B06), C human (B07), D human (B10), E *Pan paniscus* (1/97), F *Pan troglodytes* (YN 89–278), G *Gorilla gorilla* (YN 82–140), H *Pongo pygmaeus* (YN 85–38), I *Hylobates lar* (YN81–146), J *Macaca fascicularis* (11530 L). x axis codes for the normalized cortical depth (100%), y axis codes for the Grey Level Index (GLI $\pm$ SD) which is the volume proportion of cell bodies. Roman numerals indicate cortical layers. Data provided in [supplementary table S1](#).

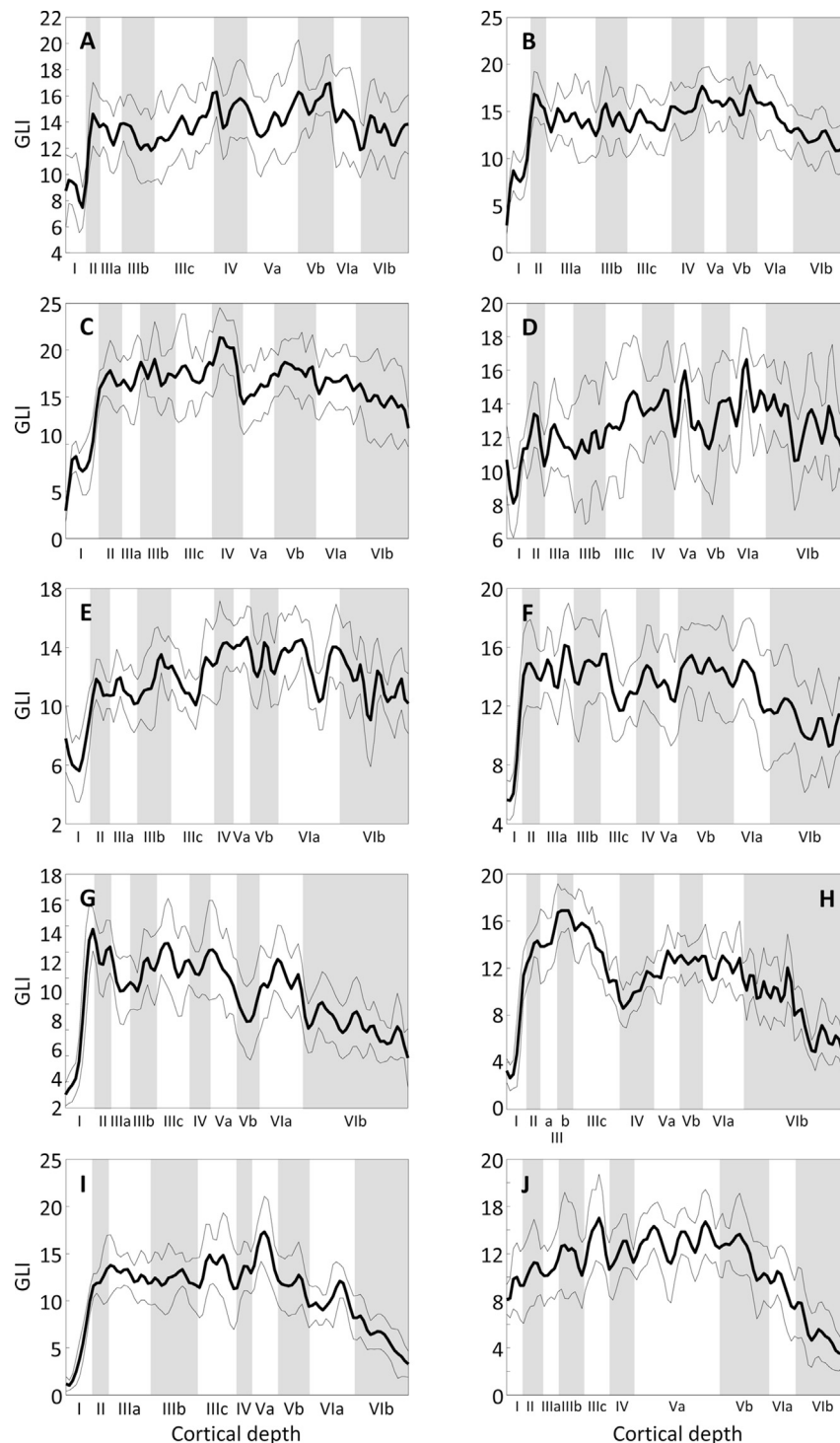


Fig. 11 – Mean GLI profiles of area 45. For further details see legend to Fig. 10. Data provided in supplementary table S2.

brains in one cluster and the gibbon and macaque brains in separate clusters. In area 4, four clusters are found, with the human brain separated from the great ape brains, the gibbon brain and the macaque brain. In areas 13 and V2, five clusters are seen. Human area 13 is separated from that of the gorilla and chimpanzee, and again separated from the orangutan and bonobo. The gibbon and macaque areas 13 are each found in a separate cluster. In area V2, the human brain is again separated from all other species. The orangutan brain is separated

from the gorilla, chimpanzee and bonobo brains. The gibbon and macaque areas V2 are located in separate clusters. Since the Euclidean distance is a measure of (dis)similarities between the cytoarchitecture of the different areas in the different species, the very small differences in area 4, 10, 13 and V2 between the great apes indicates that these three areas are very similar in cytoarchitecture in the great apes despite of the separation by the k-means clustering analysis and elbow method. Thus, the cytoarchitecture of all areas (including 44

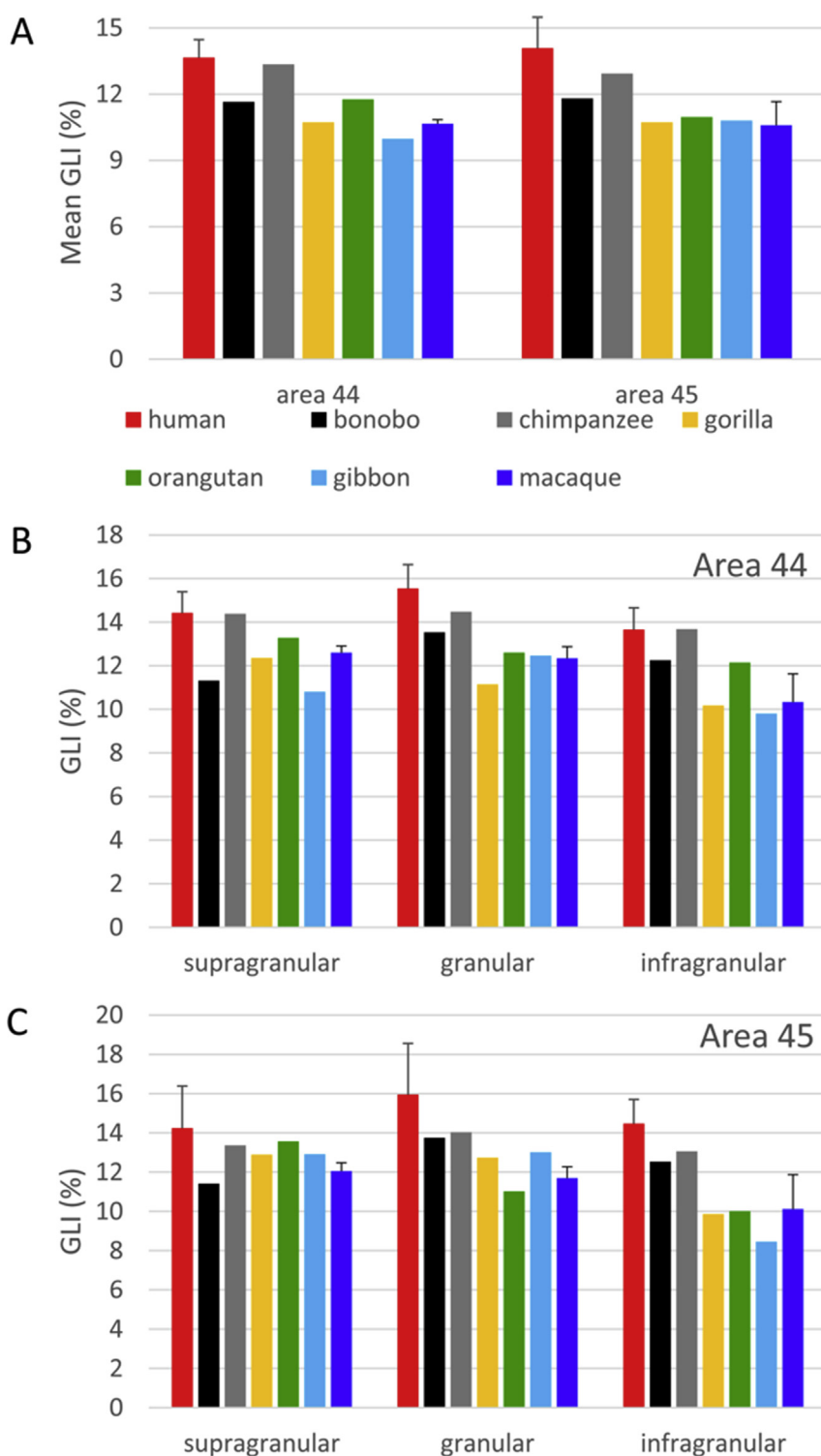


Fig. 12 – Mean ($\pm$ SD for the human and macaque data) raw GLI values (volume proportion of cell bodies per volume tissue) over all cortical layers (A) as well as in the supragranular, granular and infragranular strata of areas 44 (B) and 45 (C). Data provided in [supplementary table S3](#).

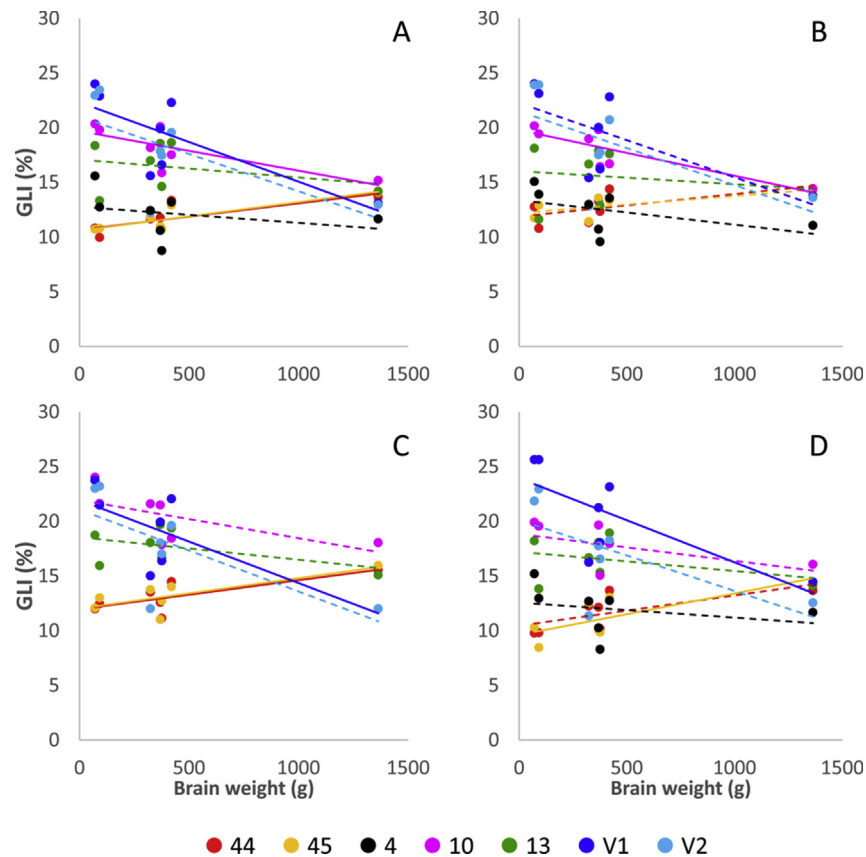


Fig. 13 – Comparison of GLI values and brain weight. **A** averaged over all cortical layers, **B** in the supragranular layers, **C** in the granular layer, **D** in the infragranular layers. Regression lines and data points of areas 44 and 45 nearly completely overlap. Continuous lines $p \leq .05$; dashed lines $p > .05$. Data for area 10 from Semendeferi et al., 2001; for area V1 and V2 from de Sousa et al., 2010 (see Material and Methods); for area 13 from Semendeferi et al., 1998; for area 4 from Sherwood et al., 2004 (macaque, orangutan, gorilla, chimpanzee, and human brains) and the present study (gibbon and bonobo brains).

and 45) in the human brain differs clearly from that of the great apes and gibbon and macaque brains.

4. Discussion

The overarching question of the present study comes from the fact that homolog areas of the human Broca region have been described in the literature mainly based on qualitative cytoarchitectonic studies, but despite the existence of homolog areas, the human language differs from that of non-human primate communication (Fitch, 2011; Fitch et al., 2010; Friederici et al., 2006a; Hauser et al., 2014, 2002; Rilling et al., 2012). However, no consistent landmarks to identify homolog cytoarchitectonic areas are present in the different species. Rather, the sulcal pattern of Broca's region largely differs between primates (Balzeau, Gilissen, Holloway, Prima, & Grimaud-Hervé, 2014; Eberstaller, 1890; Galaburda & Pandya, 1982; Keller, Roberts et al., 2009; Mingazzini, 1928; Petrides, 2013; Schenker, 2007; Schenker et al., 2010; Semendeferi et al., 2004), and the borders of these areas cannot be precisely defined by landmarks (Amunts et al., 1999;

Cantalupo & Hopkins, 2001; Keller, Crow et al., 2009; Kreht, 1936; Schenker, 2007; Schenker et al., 2010; Semendeferi et al., 2004; Sherwood et al., 2003). Therefore, we focused on the cytoarchitectonic structure of the presumable areas 44 and 45 in human and non-human primate brains and asked: what are the microstructural differences in the cytoarchitectonic organization of areas 44 and 45 between human, great ape and macaque brains which may underlie the different functionality? This approach cannot generally answer the search for microstructural differences explaining the uniqueness of human language, because future studies of temporal language-related areas are also necessary to answer this question more comprehensively. However, the Broca region seems to be a good starting point, because several publications have emphasized the similarity and comparability of areas 44 and 45 between humans, apes (Cantalupo & Hopkins, 2001; Keller, Roberts et al., 2009; Keller, Crow et al., 2009; Kreht, 1936; Schenker, 2007; Schenker et al., 2010, 2008; Semendeferi, 2004; Sherwood & Hof, 2007; Sherwood et al., 2004, 2003; Spocter et al., 2012) and macaques (Galaburda & Pandya, 1982; Morecraft et al., 2015; Petrides, 2013; Petrides & Pandya, 1994, 1999, 2002, 2009; Petrides, Cadoret, & Mackey,

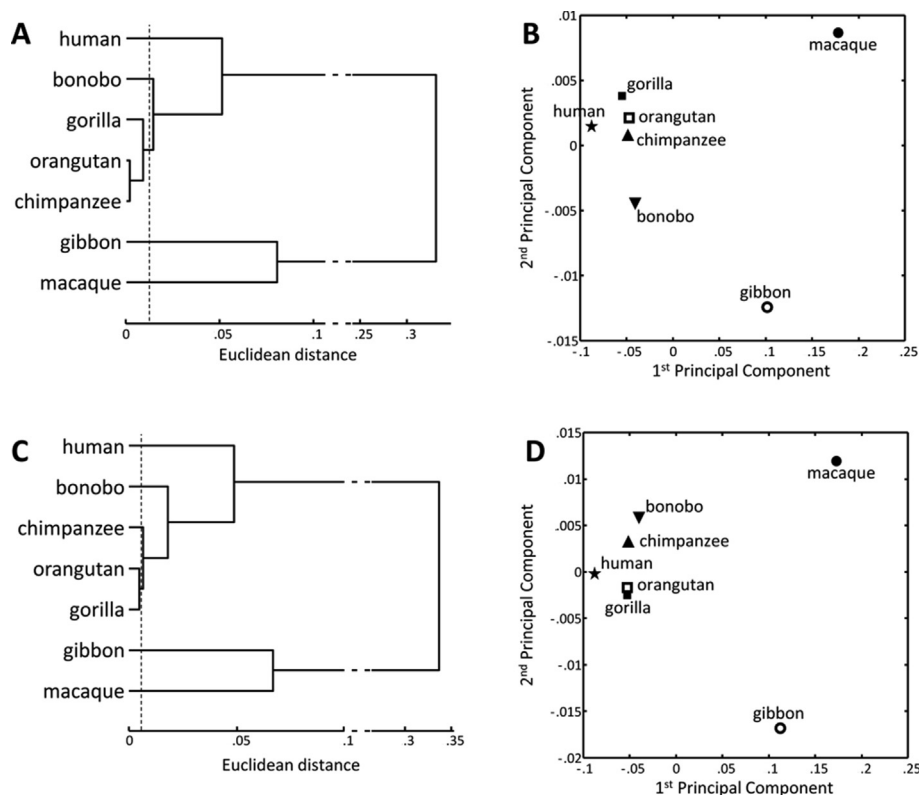


Fig. 14 – Hierarchical cluster (A, C) and principal component (B, D) analyses of the GLI values normalized by brain size and averaged over the supragranular, granular and infragranular layers of area 44 (A, B) and area 45 (C, D). Human data is represented by the mean GLI values averaged over all four human brains. Dashed lines indicate number of clusters as determined by k-means clustering and the elbow analysis.

2005, 2012) based on cytoarchitecture. The comparability is further supported by myeloarchitectonic studies (Mauss, 1908, 1911; Preuss & Goldman-Rakic, 1991; Strasburger, 1937, 1938; Vogt & Vogt, 1919), which also imply homolog areas in humans and non-human primates in the Broca region. Finally, neuroimaging and electrophysiological studies established a framework which further supports common functional aspects in the inferior ventrolateral prefrontal cortex of primates (Gil-da-Costa et al., 2006; Petrides et al., 2005; Rilling, 2014; Rilling et al., 2007; Rizzolatti & Arbib, 1998; Rizzolatti, Fogassi, & Gallese, 2001; Sherwood et al., 2004; Taglialetela et al., 2011).

As described earlier (Petrides, 2013; Petrides et al., 2005; Petrides & Pandya, 1999, 2002), we also found large pyramidal cells in layer III of area 44 which are somewhat larger than those in layer Vb; layer IV is inconspicuous. In area 45, deep layer III also contains very large pyramidal cells, even larger than those in layer V. In contrast to area 44, area 45 has well-developed layer IV.

While agreeing with these qualitatively determined similarities between human and macaque brains, the present quantitative analysis reveals differences in the proportional size of the neuropil/GLI between human and non-human primate brains. Since GLI values for the different ape species were obtained from single specimens, information concerning inter-individual variability of cytoarchitecture in ape areas 44 and 45 is not provided in this study. A comparison with data from area 10 (Semendeferi et al., 2001), 13 (Semendeferi et al., 1998), and

primary motor cortex (Sherwood et al., 2004) with the present data for areas 44 and 45 shows, however, that our single specimen analyses in apes are well within the range of cortical GLIs, and can be considered as typical for the species in question. Additionally, Spocter et al. (2012) reported neuropil fractions (neuropil fraction = $100 - \text{GLI}$) in diverse areas of the human and chimpanzee brains, which are only slightly higher than our results. The quantitative differences cannot be reduced to qualitative microscopic observations of the occurrence of large pyramidal cells in layer III, since the same cells occur in both areas and in all primate species. A quantitative analysis of cytoarchitecture is required, because the differences may be caused by a layer- and area specific development of the neuropil between species. Thus, the analysis of the relation between the size of the neuropil and the space occupied by cell bodies in and between layers and areas can highlight a major aspect of cytoarchitecture in general. Classical cytoarchitectonic publications (Brodman, 1909; von Economo & Koskinas, 1925) had the differences in cell packing density between the layers of different areas in the focus of their work, but they contain only qualitative estimates (Brodman, 1909) or cell counting using biased methods (von Economo & Koskinas, 1925). Unbiased sampling and measuring methods were not available at this time. Since the neuropil comprises afferent and efferent connections as well as dendrites and synapses, differences in microcircuitry and connectivity can be assumed, if species and intraspecies areal and laminar differences are found. The present study provides measurements of the

volume proportions of the neuropil and the cell bodies which reveals differences in the microscopical dimensions of both tissue compartments. The human brain shows the expected significantly larger volume proportions of neuropil in areas V1 (averaged over all layers, granular and infragranular layers) and 10 (averaged over all layers, supragranular layers), and non-significant trends in areas 4, 13, and V2 when compared with apes and macaque. Contrastingly, the neuropil proportion in areas 44 and 45 is significantly lower, or shows a trend to lower values, in humans than in apes or macaque brains. Thus, as shown by the hierarchical cluster analyses, a partitioning of the interspecies differences was found which sets the cytoarchitecture of the human Broca areas apart from that of non-human primates. The great apes also differ considerable in cytoarchitecture from gibbon and macaques. The difference in cytoarchitecture between the great apes is very small, with the exception of the bonobo, but here only in the Broca areas. In all other areas of the frontopolar, orbitofrontal, motor and visual cortices the cytoarchitecture of the human brain is similar to

that of the great apes (particularly in areas 10 and V1), or shows only minor differences. Only the cytoarchitecture of gibbon and macaque brains is consistently different from that of human and great ape brains. Thus, we suggest that areas 44 and 45 of the Broca region, which are involved in the language system (Amunts & Zilles, 2006, 2012; Amunts et al., 2010, 2004; Bookheimer, 2002; Fink et al., 2006; Friederici, 2017a; Friederici et al. b, 2006a; Horwitz et al., 2003; Manjaly, Marshall, Stephan, Gurd, Zilles, & Fink, 2005; Turken & Dronkers, 2011; Zilles & Amunts, 2018), may evolve their neuropil fraction following a different developmental trajectory. They increase in GLI and decrease in neuropil fraction compared with all smaller brains of non-human primates, whereas all other cortical areas studied here display an opposite trend.

These cytoarchitectonic observations in the Broca region are further supported by evolutionary differences in the arcuate and uncinate fascicles, which differentially target areas 44 and 45 (Rilling et al., 2008, 2012). In fact, differences between human, great ape and macaque brains are also found

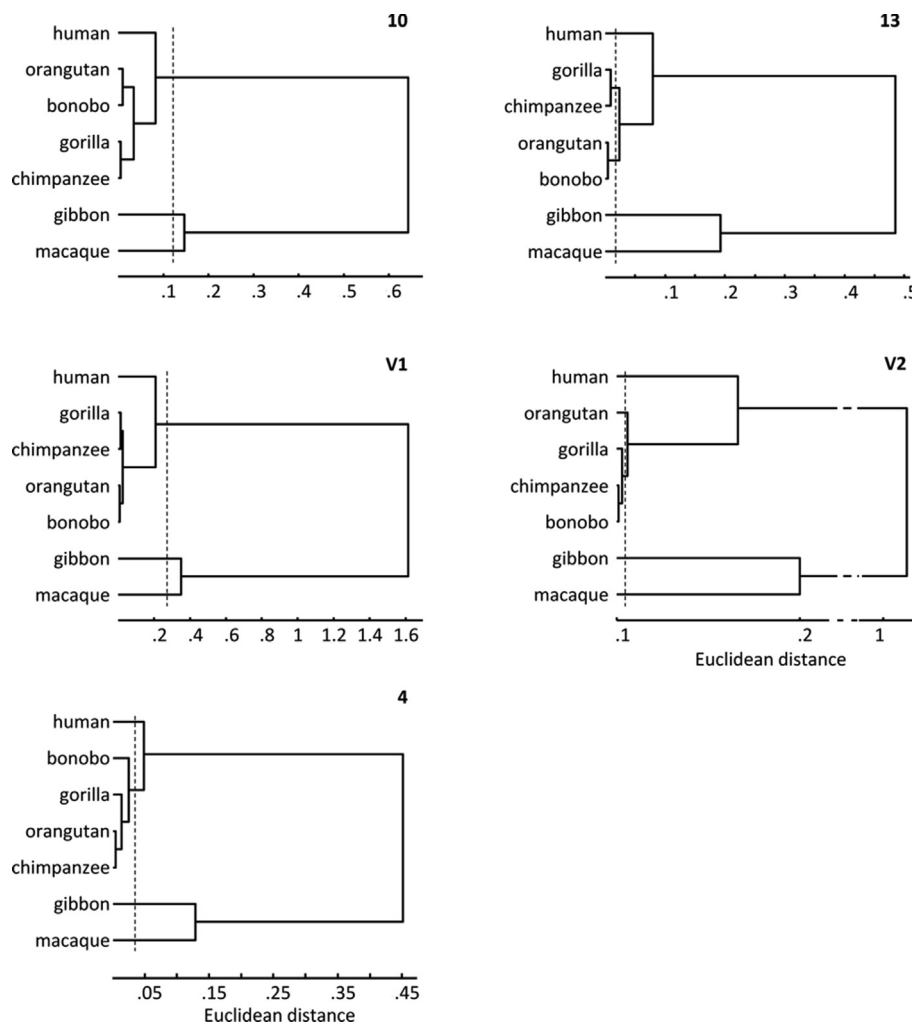


Fig. 15 – Hierarchical cluster analyses of the GLI values normalized by brain size and averaged over the supragranular, granular and infragranular layers of areas 10 (frontopolar area), 13 (part of the orbitofrontal cortex), V1 (primary visual area), V2 (secondary visual area), and 4 (primary motor area). Dashed lines indicate the number of clusters as determined by the k-means and elbow analyses. Data for area 10 from Semendeferi et al., 2001; for area V1 and V2 from de Sousa et al., 2010; for area 13 from Semendeferi et al., 1998; for area 4 from Sherwood et al., 2004 (macaque, orangutan, gorilla, chimpanzee, and human brains) or from the present study (gibbon and bonobo brains).

in the size and lateralization of these two major afferent fiber tracts from the temporal language region (Frey et al., 2008; Friederici, 2017a, b; Friederici et al., 2006a; Jbabdi et al., 2013; Petrides, 2013; Petrides & Pandya, 2002, 2009; Rilling et al., 2012; Watanabe-Sawaguchi et al., 1991). This is particularly true for the more dorsally located arcuate fascicle, which is thought to have undergone greater evolutionary changes than the uncinate fascicle to enable the processing of verbal information in humans (Kelly et al., 2010; Rilling et al., 2008). Although the spacing between cortical minicolumns caused by neuropil-rich septa is wider in human areas 44 and 45 than in non-human primates (Rilling, 2014; Schenker et al., 2008), the larger GLI value in human compared to non-human primate brains may be caused by a considerable increase in neuronal density in the minicolumns.

If the GLIs of areas 4, 10, 13, 44, 45, V1 and V2 are correlated with brain weight, positive correlations were found for areas 44 and 45 of the Broca region (area 44: $p = .05$; area 45: $p = .01$). In contrast, negative correlations were found for areas V1 and 10 ($p = .05$). The other areas show trends to negative correlations, but did not reach the level of significance ($p > .05$). That means, that areas 44 and 45 show an increasing density of the volume proportion of all cell bodies with increasing brain weight, whereas the other areas show significant, or non-significant trends to an increasing volume proportion of the neuropil which is the space containing myelinated and unmyelinated nerve fibers, dendrites, processes of glial cells and blood vessels. Thus we conclude, that with increasing brain size also the number of neurons increases since a tight correlation between GLI and number of cells has been demonstrated despite of varying cell sizes (Wree et al., 1982). We could not use the data for visual areas V1 and V2 as reported in the literature (de Sousa et al., 2010) because they are by far higher than any GLI measurement in numerous cortical areas. The reason for this discrepancy is a methodical difference between the V1/V2 data and all other data. The V1/V2 data are not measured like all the other GLIs. de Sousa et al. (2010) did not convert the grey values in their 8-bit GLI images (i.e., from 0 to 255) into GLI values from 0% to 100%. Therefore, they reported unusually high “GLI values” which had to be corrected. Also the original data of the study on areas 44 and 45 by Schenker et al. (2008) was not used for the correlation analysis between GLI and brain weight, because they are based on a completely different measuring technique (photographs of the histological sections followed by a densitometric analysis; different criteria for the segmentation of cell bodies from the background).

5. Conclusions

The present paper is a quantitative approach for defining differences in microstructure, particularly of areas 44 and 45, in human and non-human primate brains. The GLI profiles through these areas show different distribution patterns between the species, and were used to define the cell-body (GLI) and neuropil fraction in supragranular, granular and infragranular layers. Correlation and multivariate analyses demonstrated that areas 44 and 45 have a different trajectory of the volume proportion between cell bodies and neuropil when these areas are compared with the homolog areas in

non-human primate brains. Furthermore, the multivariate analyses showed that the cytoarchitecture of the human brain differs considerably from that of the non-human primates, particularly the gibbon and macaque brains. Interestingly, the bonobo brain has the most similar cytoarchitecture compared with the human brain, but only in areas 44 and 45, and not in frontopolar area 10, orbitofrontal area 13, motor area 4 or visual areas V1 and V2. Future studies including the temporal language region are necessary to analyze additional aspects of the microcircuitry and molecular organization of language areas in primate brains to further elucidate the unique neuronal basis of the language faculty in humans.

Acknowledgements

This project was partially funded by the European Union's Horizon 2020 Research and Innovation Programme under Grant Agreement No. 785907 (HBP SGA2).

Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.cortex.2018.09.008>.

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